Confounders and controls

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete intended-factor and alternative-explanation record.

Keep controls, condition cells, object IDs, observed/planned status and causal limits together.

Keep one habit: Recall → W1–W3/W5 → independent timing design → confound/control project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Experimental units and independence. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Methods results and limitations. Give an example and a mistake that the idea helps you avoid.

Experimental units and independence

My remembered idea, example and correction

Hypotheses and measurable outcomes

My remembered idea, example and correction

Methods results and limitations

My remembered idea, example and correction

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A confounder changes with the intended factor and offers another explanation; controls hold relevant conditions fixed or vary them systematically. Main A uses c1/Q1–Q3 while B uses c2/Q4–Q6, alongside the method change. Proposed matched comparison keeps corpus, queries, cutoff 3, rubric r1 and toy-local environment comparable; no new outcomes or actual rerun are supplied.

  Matching listed fields is not proof every alternative was removed. Separate picture couples healthy leaves with white paper and spotted leaves with blue. Cross healthy-white, healthy-blue, spotted-white and spotted-blue. A deliberately paper-only rule can fit original two records yet score two of four crossed records, illustrating a possible shortcut rather than diagnosing a real model.

  Same held-out leaf across backgrounds is a repeated diagnostic view, not another independent plant; preserve all object views outside training for a new-object goal. Changed-background sensitivity does not reveal complete internal reasoning. Code/diagram are fictional metadata/illustrations with no actual photos, training, randomization, collection or timing experiment.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- A confounder changes with the intended factor and offers another explanation; controls hold relevant conditions fixed or vary them systematically. Main A uses c1/Q1–Q3 while B uses c2/Q4–Q6, alongside the method change. Proposed matched comparison keeps corpus, queries, cutoff 3, rubric r1 and toy-local environment comparable; no new outcomes or actual rerun are supplied.

  Matching listed fields is not proof every alternative was removed. Separate picture couples healthy leaves with white paper and spotted leaves with blue. Cross healthy-white, healthy-blue, spotted-white and spotted-blue. A deliberately paper-only rule can fit original two records yet score two of four crossed records, illustrating a possible shortcut rather than diagnosing a real model.

  Same held-out leaf across backgrounds is a repeated diagnostic view, not another independent plant; preserve all object views outside training for a new-object goal. Changed-background sensitivity does not reveal complete internal reasoning. Code/diagram are fictional metadata/illustrations with no actual photos, training, randomization, collection or timing experiment.

- New method uses better corpus and different queries while old method uses the original set. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A confounder changes with the intended factor and offers another explanation; controls hold relevant conditions fixed or vary them systematically. Main A uses c1/Q1–Q3 while B uses c2/Q4–Q6, alongside the method change. Proposed matched comparison keeps corpus, queries, cutoff 3, rubric r1 and toy-local environment comparable; no new outcomes or actual rerun are supplied.

  Matching listed fields is not proof every alternative was removed. Separate picture couples healthy leaves with white paper and spotted leaves with blue. Cross healthy-white, healthy-blue, spotted-white and spotted-blue. A deliberately paper-only rule can fit original two records yet score two of four crossed records, illustrating a possible shortcut rather than diagnosing a real model.

  Same held-out leaf across backgrounds is a repeated diagnostic view, not another independent plant; preserve all object views outside training for a new-object goal. Changed-background sensitivity does not reveal complete internal reasoning. Code/diagram are fictional metadata/illustrations with no actual photos, training, randomization, collection or timing experiment.

- A report omits this qualification: **Control**s address specified alternatives, not every possible unknown **confounder**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- A confounder changes with the intended factor and offers another explanation; controls hold relevant conditions fixed or vary them systematically. Main A uses c1/Q1–Q3 while B uses c2/Q4–Q6, alongside the method change. Proposed matched comparison keeps corpus, queries, cutoff 3, rubric r1 and toy-local environment comparable; no new outcomes or actual rerun are supplied.

  Matching listed fields is not proof every alternative was removed. Separate picture couples healthy leaves with white paper and spotted leaves with blue. Cross healthy-white, healthy-blue, spotted-white and spotted-blue. A deliberately paper-only rule can fit original two records yet score two of four crossed records, illustrating a possible shortcut rather than diagnosing a real model.

  Same held-out leaf across backgrounds is a repeated diagnostic view, not another independent plant; preserve all object views outside training for a new-object goal. Changed-background sensitivity does not reveal complete internal reasoning. Code/diagram are fictional metadata/illustrations with no actual photos, training, randomization, collection or timing experiment.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Every method A run is morning and B run afternoon; network speed varies by time. Propose a repair.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled **confounder**s and **control**s **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **confounder**s and **control**s.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Confounder**s and **control**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Confounder**s and **control**s. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Every method A run is morning and B run afternoon; network speed varies by time. Propose a repair.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

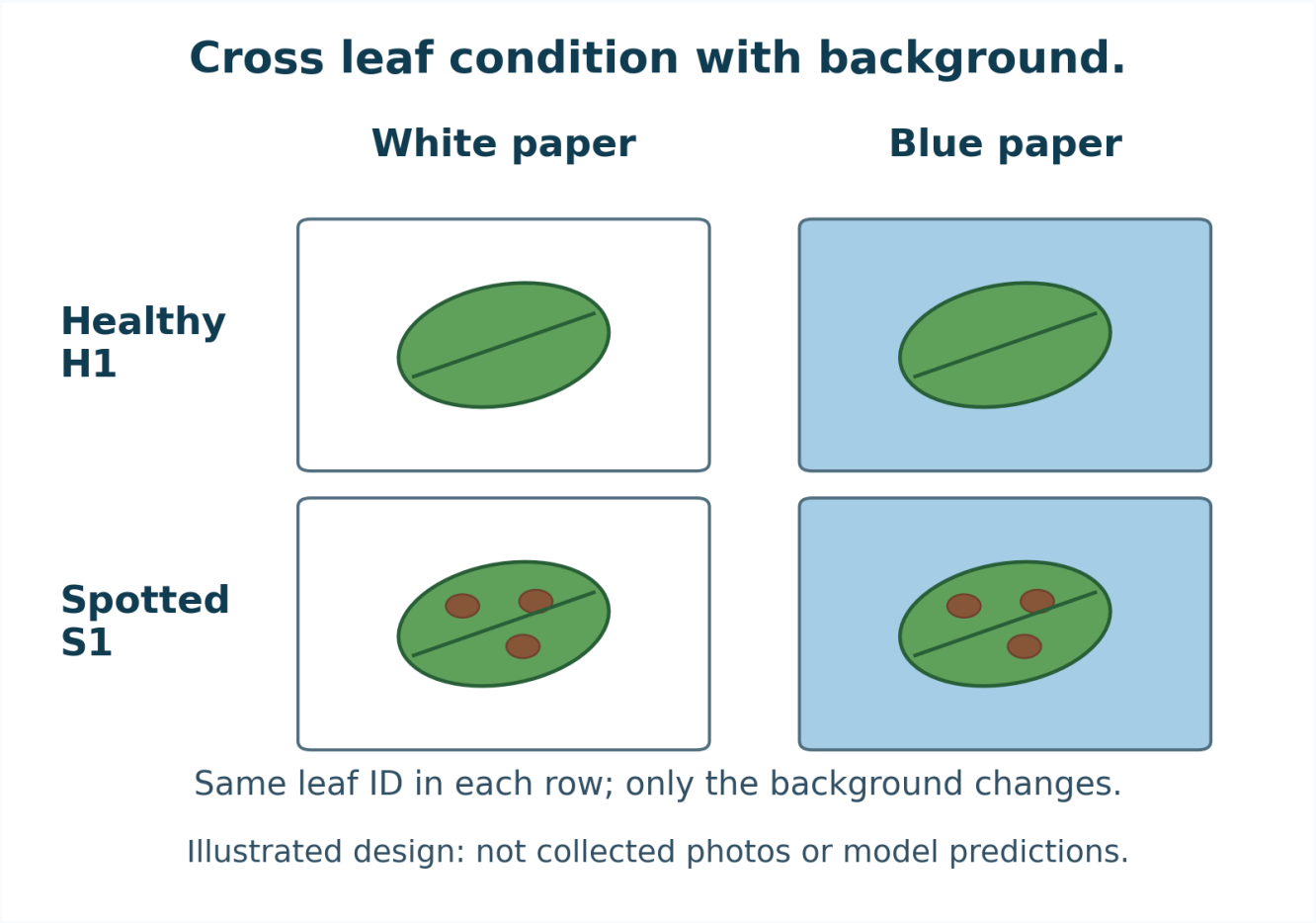
- Separate background from class. **Training** data show every healthy leaf on white paper and every spotted leaf on blue paper.

- Goal is leaf condition, not paper colour. A proposed check uses both paper colours for each condition.

- Draw the four condition/background combinations. Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/10

Cross class with paper background

	White paper	Blue paper
Healthy	Collect	Collect
Spotted	Collect	Collect

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/10

Confounders and controls

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify coupled variables

Condition and background change together in the original collection.



2. Cross the conditions

Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases.



3. Inspect the shortcut

Compare predictions when the same leaf's background changes under a controlled test.

CHANGE ONE THING

New cases still use only healthy-white and spotted-blue.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Separate background from class. **Training** data show every healthy leaf on white paper and every spotted leaf on blue paper.
- Which labelled step matches this detail? Compare **prediction**s when the same leaf's background changes under a **control**led test.
- 1. Identify coupled variables
- 2. Inspect the shortcut
- 3. Cross the conditions
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Inspect the shortcut Compare **prediction**s when the same leaf's background changes under a **control**led test.

A **confounder** can provide an alternative explanation for a result. Crossing conditions makes the intended feature and the nuisance feature less entangled.

Changed-case question and optional hint

- Changed-case practice: New cases still use only healthy-white and spotted-blue. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. The collection does not resolve the background confounding.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The collection does not resolve the background confounding. The same two variables remain coupled instead of being varied separately.

Repair and retain the record

Review Experimental units and independence (ai-7/9). Goal: Explain the **mechanism** and terminology of experimental units and independence. Short practice: Count independent plants and photo views.

Construct the stated group-correct split, then explain why placing a P1 view in evaluation creates leakage for the new-plant goal.. Return to this week and explain what you changed.

Review Hypotheses and measurable outcomes (ai-7/7). Goal: Explain the **mechanism** and terminology of hypotheses and measurable outcomes. Short practice: Name the predicted outcome and how unsupported is scored.

If B sounds friendlier but ties A's unsupported count, does the recorded improvement hypothesis receive support?. Return to this week and explain what you changed.

Review Methods results and limitations (ai-7/4). Goal: Explain the **mechanism** and terminology of methods results and limitations.

Short practice: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the result establish that B is best for outdoor farming?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

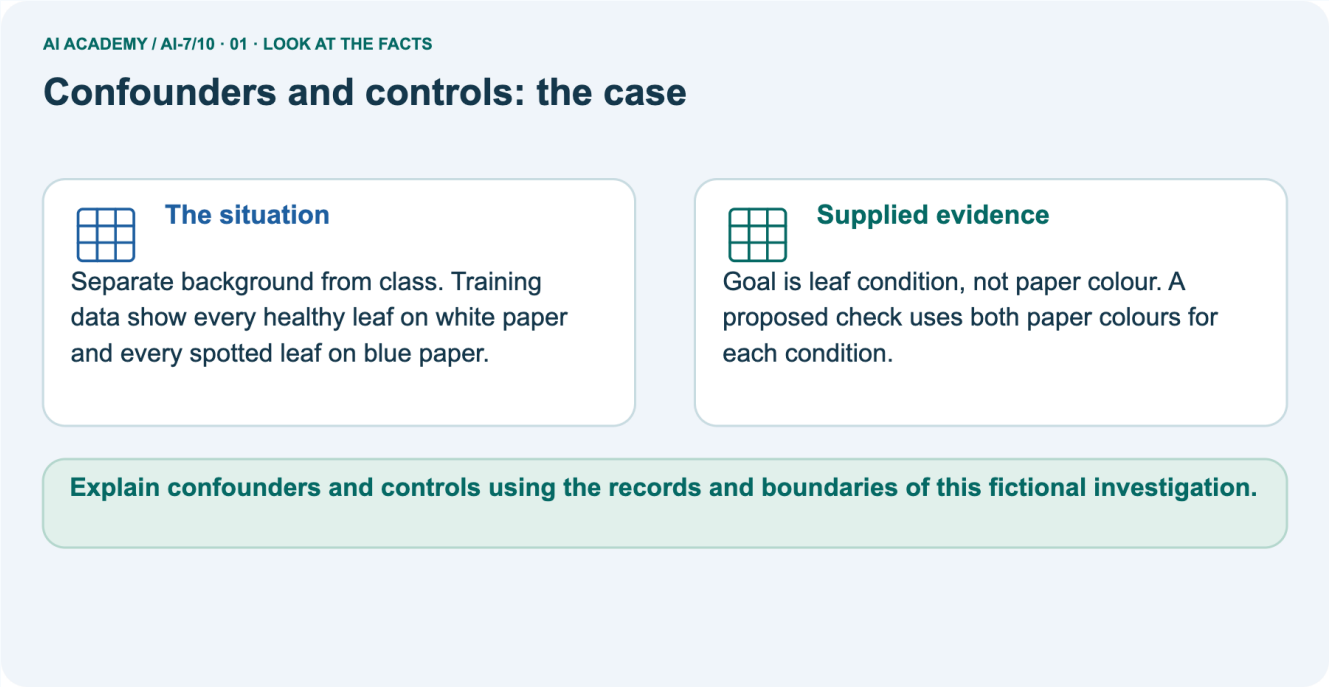
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Confounders and controls: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

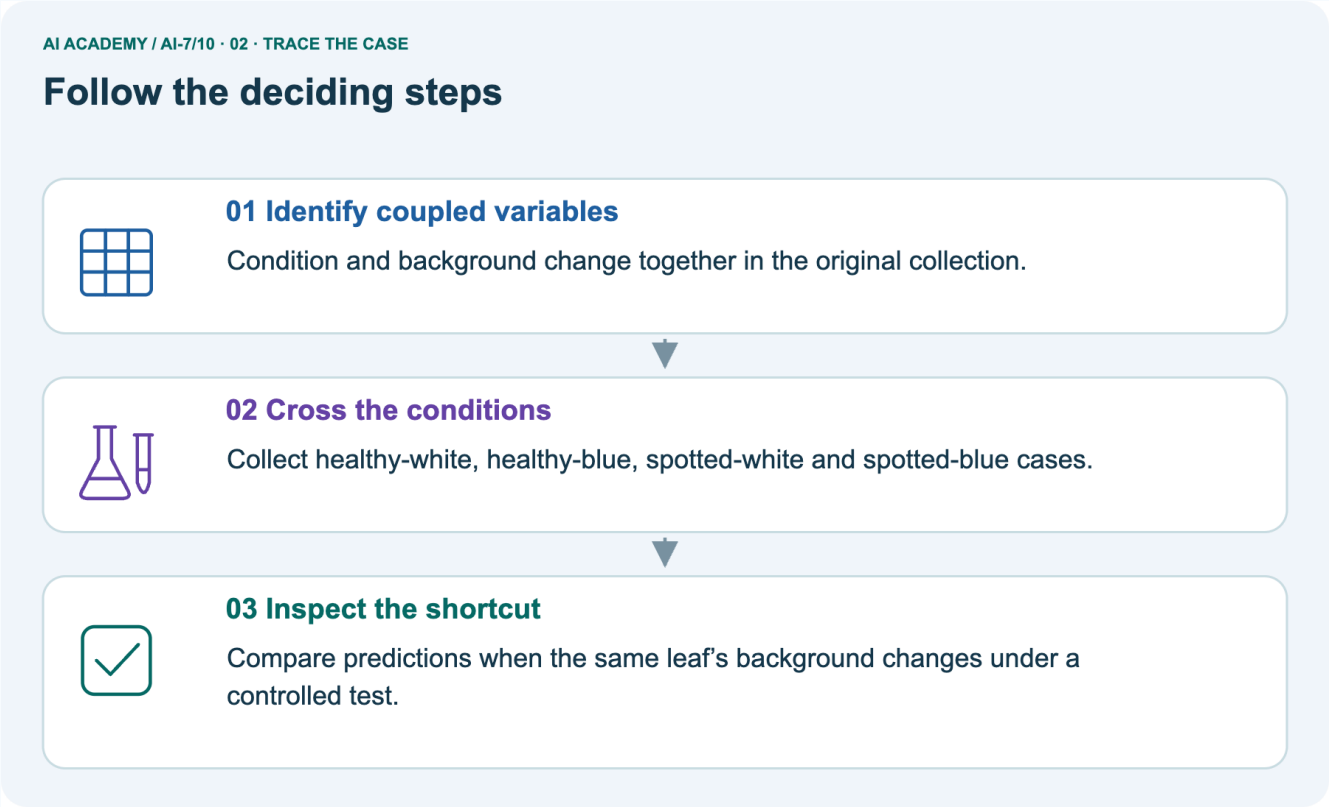
Separate background from class. **Training** data show every healthy leaf on white paper and every spotted leaf on blue paper. Supplied facts: Goal is leaf condition, not paper colour.

A proposed check uses both paper colours for each condition.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



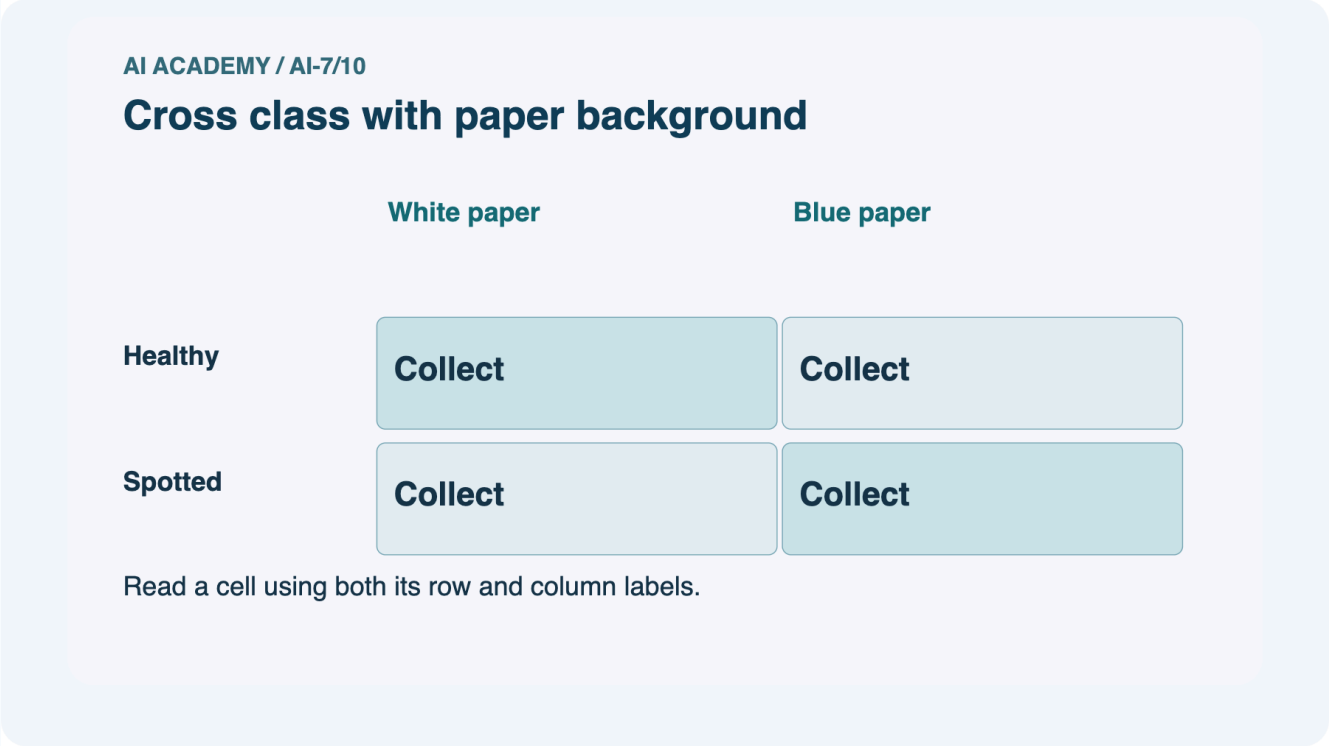
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify coupled variables: Condition and background change together in the original collection. → Cross the conditions: Collect healthy-white, healthy-blue, spotted-white and spotted-blue cases. → Inspect the shortcut: Compare **prediction**s when the same leaf's background changes under a **control**led test.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Cross class with paper background



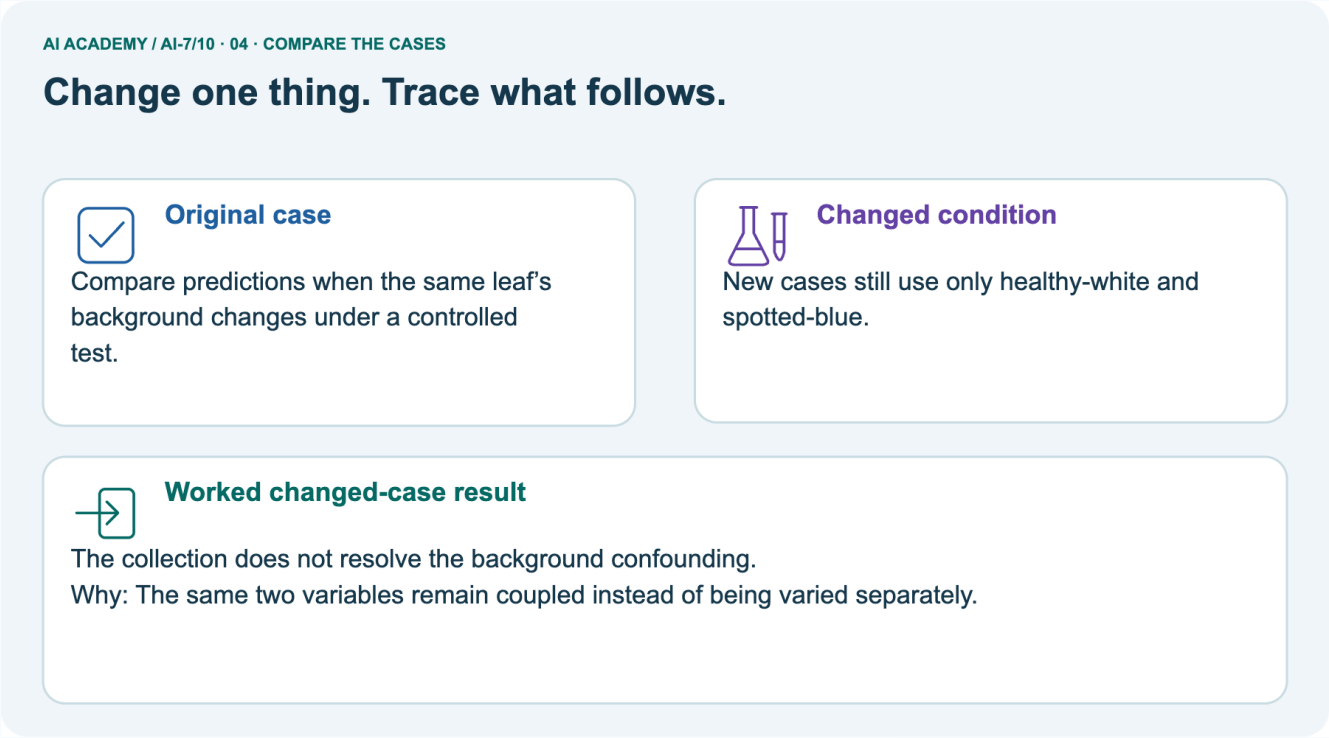
Follow the labelled picture with the matching explanation. Open the vector diagram

Cross class with paper background. Healthy: White paper = Collect, Blue paper = Collect. Spotted: White paper = Collect, Blue paper = Collect.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

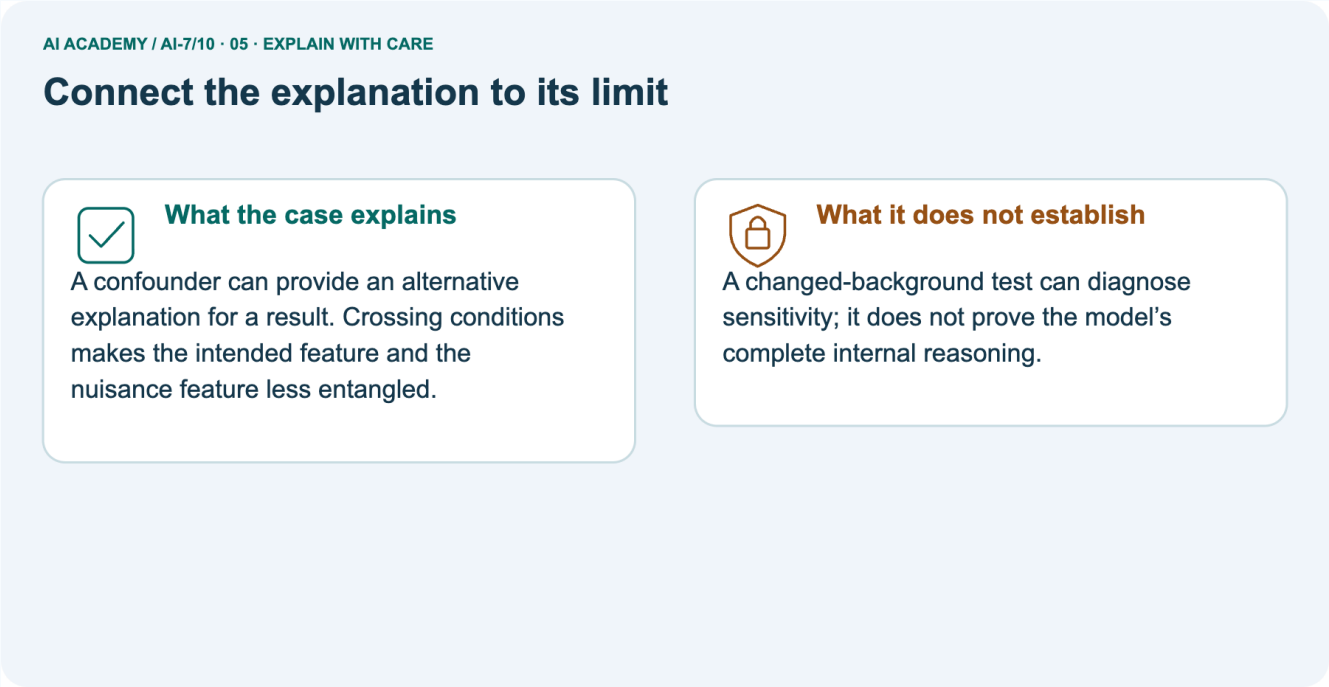
Original case: Compare **prediction**s when the same leaf's background changes under a **control**led test. Changed condition: New cases still use only healthy-white and spotted-blue.

Expected result: The collection does not resolve the background confounding. Reason: The same two variables remain coupled instead of being varied separately.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

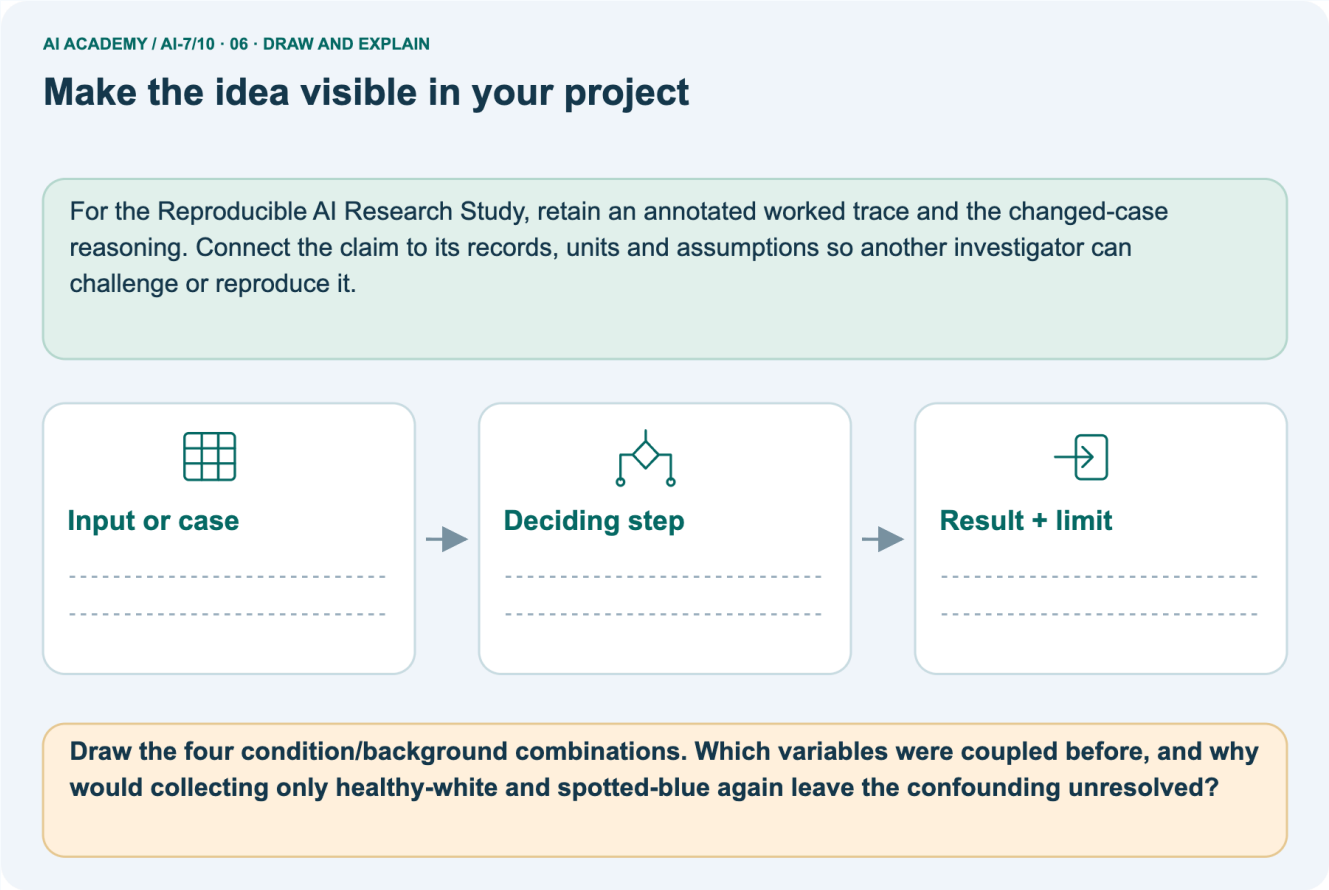
A **confounder** can provide an alternative explanation for a result. Crossing conditions makes the intended feature and the nuisance feature less entangled.

Limit: A changed-background test can diagnose sensitivity; it does not prove the model's complete internal reasoning.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Draw the four condition/background combinations.

Which variables were coupled before, and why would collecting only healthy-white and spotted-blue again leave the confounding unresolved?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Confounders and controls: the case

02. Follow the deciding steps

03. Cross class with paper background

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.